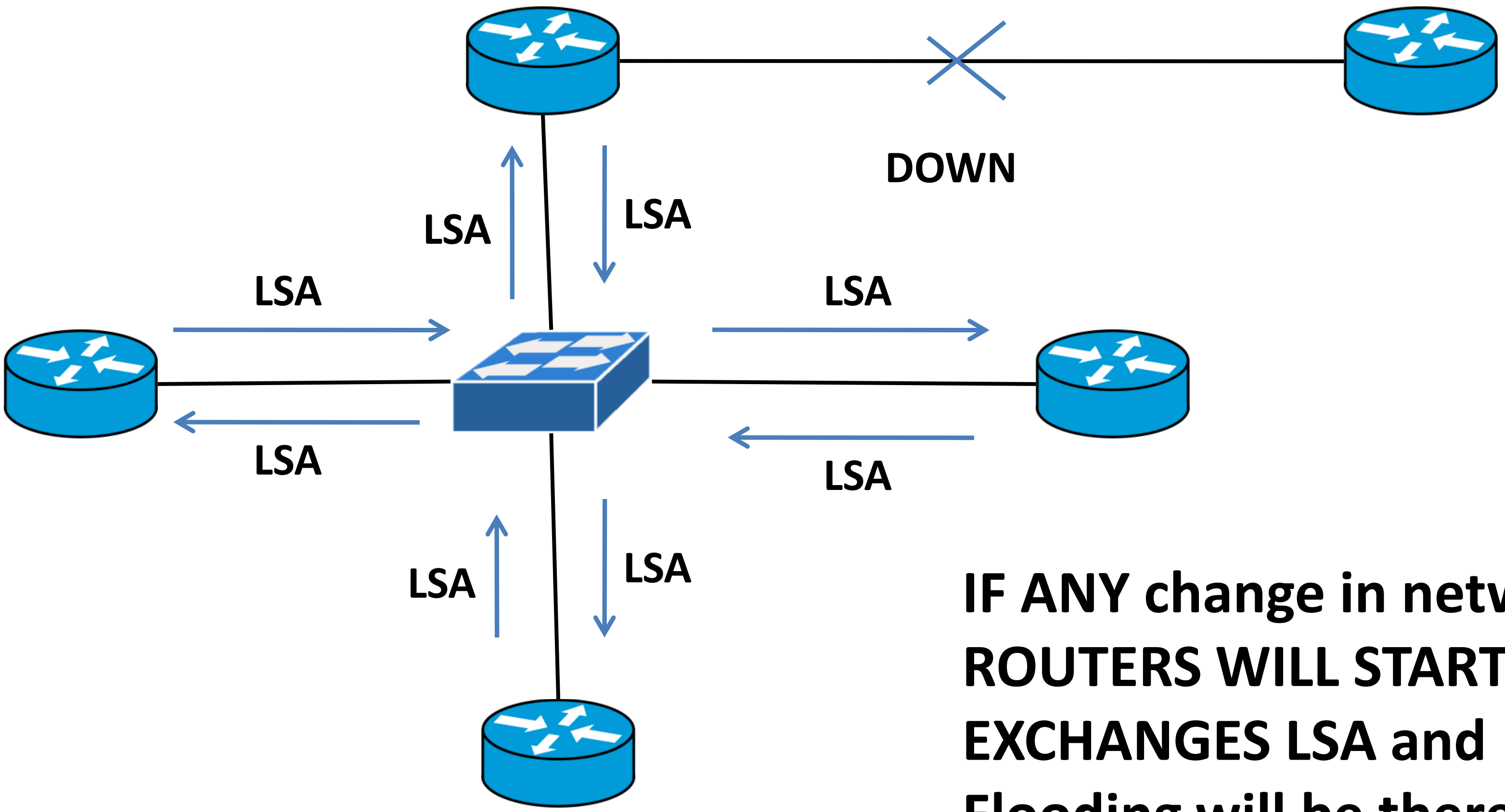
<u>OSPF LSA FLOODING:</u>

LSA (Link-State Advertisement) Flooding is a crucial mechanism in the OSPF (Open Shortest Path First) protocol that ensures all routers within an OSPF area have a consistent view of the network topology. The process involves the distribution of LSAs throughout the OSPF network to inform routers of network changes, such as new links, removed links, or changes in link metrics.

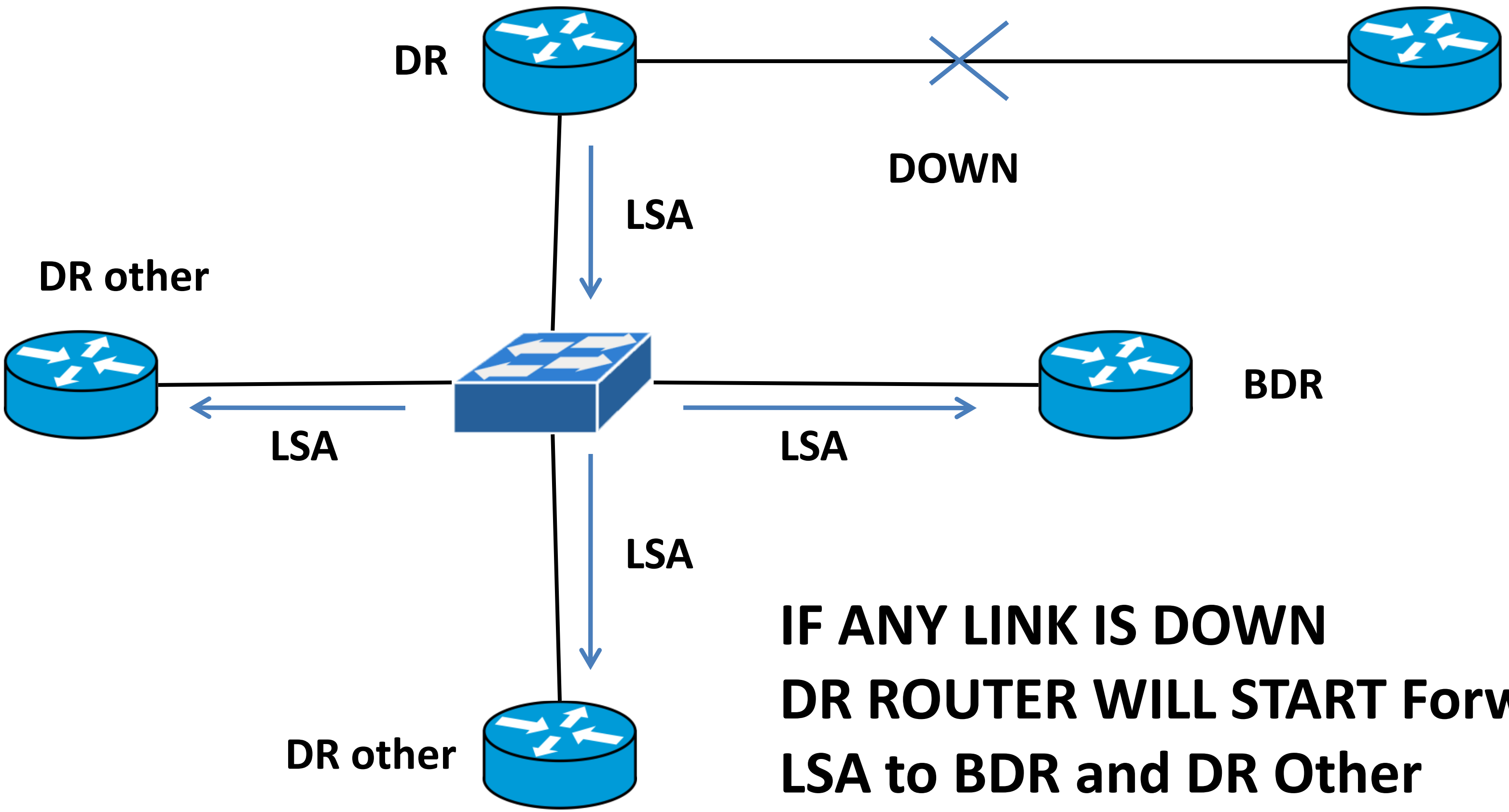
<u>Initial Flooding</u>: When a router detects a change in the network (like an interface going up or down), it generates an LSA and floods it out to all of its OSPF neighbors on all interfaces.

Without DR/BDR ELECTION:



IF ANY change in networks ROUTERS WILL START EXCHANGES LSA and Flooding will be there

After DR/BDR ELECTION:



IF ANY LINK IS DOWN
DR ROUTER WILL START Forwarding
LSA to BDR and DR Other

DR/BDR Election Procedure

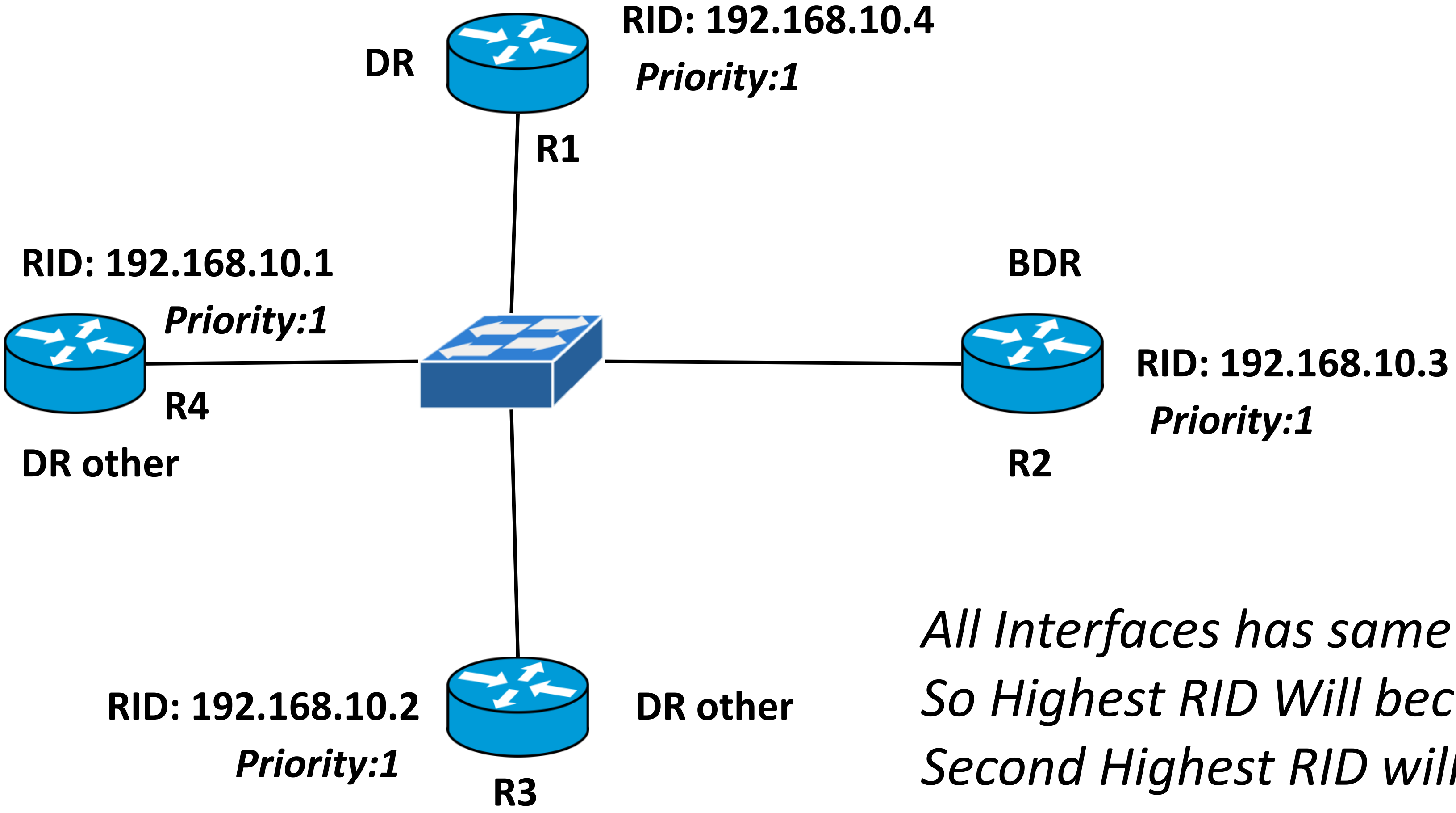
Router Priority: Each OSPF router interface has a configurable priority value ranging from 0 to 255. The router with the *highest priority on a segment is elected as the DR.*

Default Priority: The default OSPF priority is 1.

Priority 0: If a router’s interface has a priority of 0, it is ineligible to become the DR or BDR.

Router ID: If two or more routers have the same priority, *the router with the highest Router ID is elected as the DR.*.

Example of DR, BDR & DR Other Election:



All Interfaces has same Priority
So Highest RID Will become DR
Second Highest RID will be BDR

Commands

Check OSPF Neighbor Details (DR and BDR Status)

Router>enable

Router#show ip ospf neighbor

Changing the Priority Number:

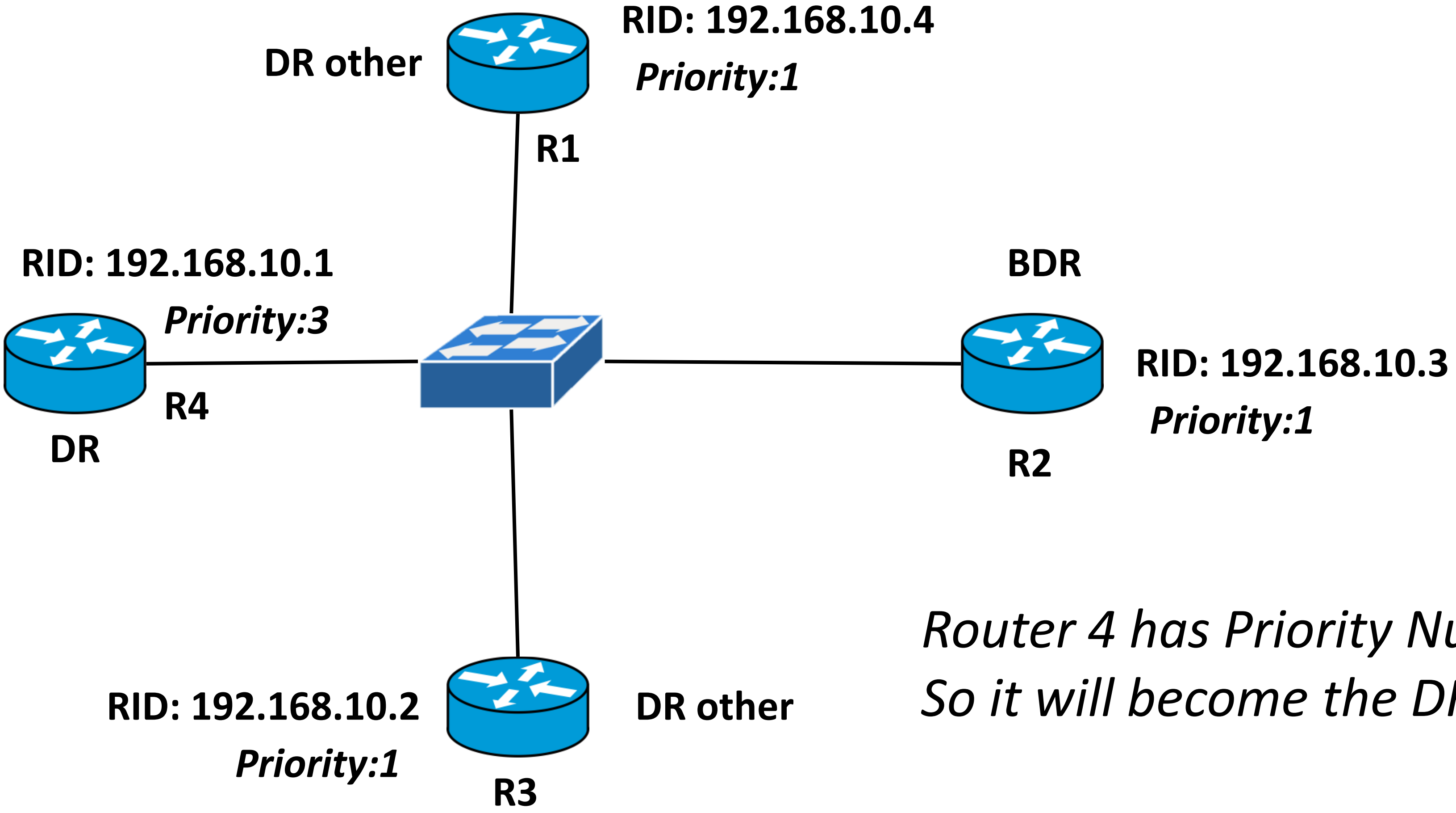
Router>enable

Router#Conf t

Router(config)#int g0/0/0

Router(config-if)#ip ospf priority 3

Example of DR, BDR & DR Other Election:



Router 4 has Priority Number 3
So it will become the DR